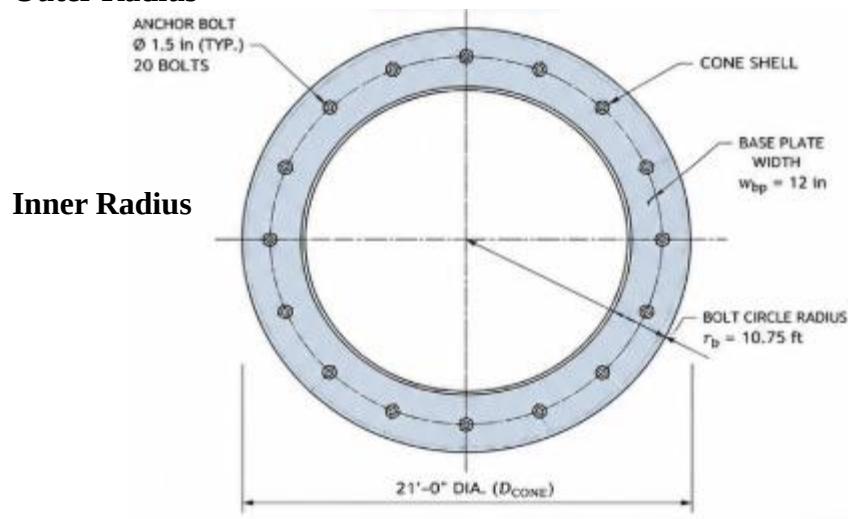


2. Base plate segment

Parameter	Symbol	Value	Units
Cone diameter	D_{cone}	21	ft
Width of ring wall	W_{rw}	2	ft
Number of anchor bolts	N_b	20	—
Anchor bolt diameter	d_b	1.5	in
Design Strip	N	12	in
Vertical load on foundation	P	1650.2553	kips
Governing overturning moment	M_u	4847.67	kip-ft
Steel yield strength	F_y	36	ksi
Steel unit weight	γ_s	490	pcf
Number of plate segments	N_s	4	—
LRFD resistance factor	ϕ	0.90	—
Modulus of elasticity	E	29000	ksi

1.1. Geometry Inputs Derived from These:

Outer Radius

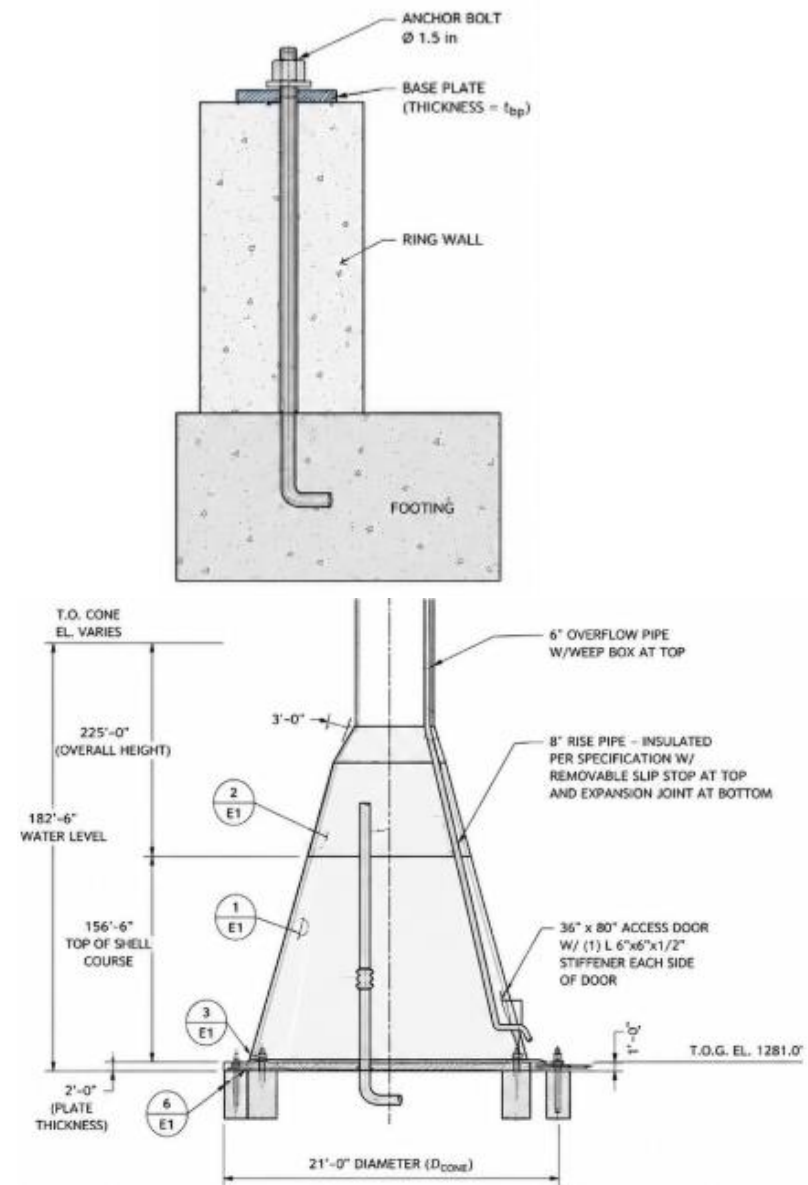


$$R_o = \frac{D_{cone}}{2} + \frac{w_{rw}}{4}$$

$$R_o = 11ft$$

$$R_i = \frac{D_{cone}}{2} - \frac{w_{rw}}{4}$$

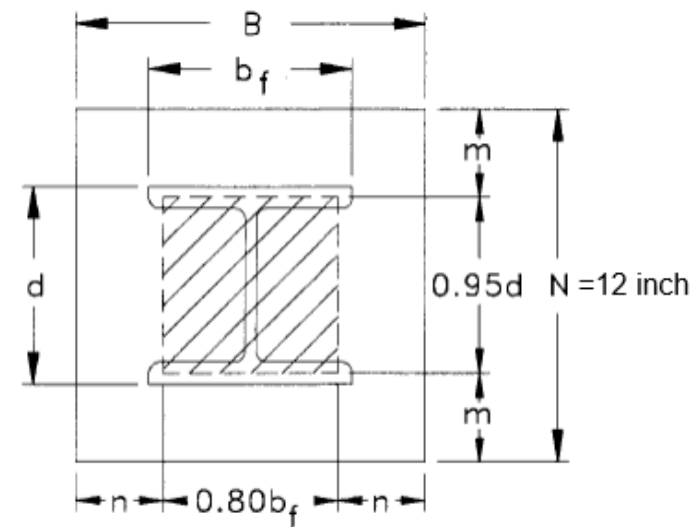
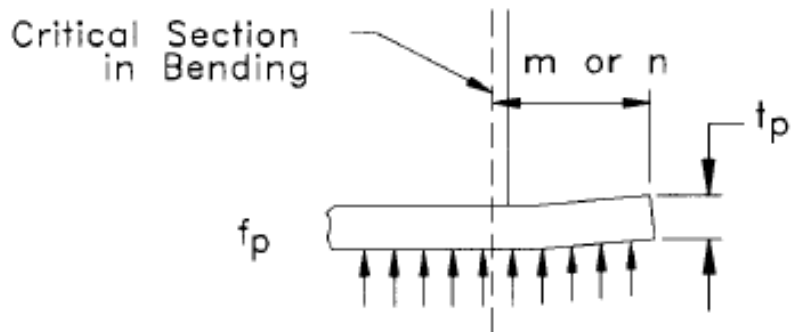
$$R_i = 10ft$$



1.2. Annular Plate Width Based on Ring Geometry

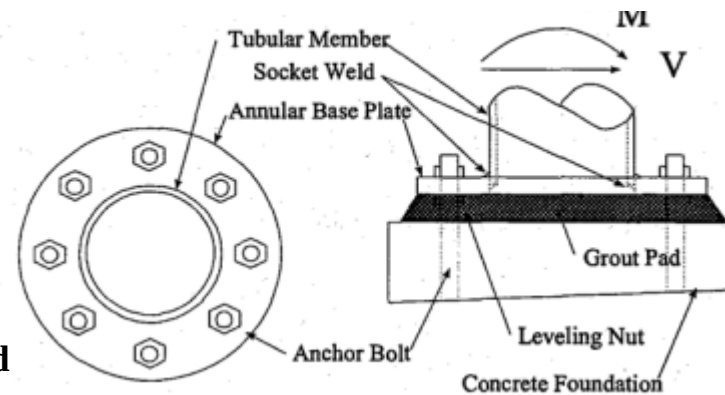
The annular radial plate width may also be expressed as:

$$B = R_o - R_i = 11 - 10 = 1 \text{ ft} = 12 \text{ in}$$



(b) Critical Sections

Annular base plate with grout pad



Step 2: Design against Axial Load

2.1. Convert Total Load to Circumferential Load

Original AISC Procedure

Annular Modification: Use load per circumferential length: $w_u = \frac{P_u}{\pi D_{cone}}$

Substitute:

$$w_u = \frac{1650.2553}{\pi(21)}$$

$$w_u = \frac{1650.2553}{65.973}$$

$$w_u = 25.01 \text{ kips/ft}$$

Convert to kips/in:

$$w_u = \frac{25.01}{12}$$

$$w_u = 2.084 \text{ kips/in}$$

$$w_u = \frac{1650.2553}{\pi(21)} = \mathbf{25.01 \text{ kips/ft}}$$

2.2. Define Design Strip Width

This represents a 1-ft circumferential strip.

$$N = \mathbf{12 \text{ in}}$$

2.3. Determine Load on One Strip

Modified Strip Load

$$P_{strip} = w_u(N)$$

Substitute:

$$P_{strip} = 2.084(12)$$

$$P_{strip} = 25.01 \text{ kips}$$

$$P_{strip} = 2.084(12) = \mathbf{25.01 \text{ kips}}$$

2.4. Determine Required Plate Area

Modified Equation

$$A_1 = \frac{P_{strip}}{1.7\phi_c f'_c}$$

Substitute:

$$A_1 = \frac{25.01}{1.7(0.65)(4)}$$

$$A_1 = \frac{25.01}{4.42}$$

$$A_1 = 5.66 \text{ in}^2$$

$$A_1 = \frac{25.01}{1.7(0.65)(4)} = \mathbf{5.66 \text{ in}^2}$$

2.5. Plate Dimension Equations

Equation for N

$$N = \sqrt{A_1} + \Delta$$

$$N = \sqrt{A_1} + \Delta$$

Equation for Δ

$$\Delta = 0.50(0.95d - 0.80b_f)$$

$$\Delta = 0.50(0.95d - 0.80b_f)$$

2.6. Compute Δ

For annular strip approximation:

$$d = 12 \text{ in}$$

$$b_f = 12 \text{ in}$$

Substitute:

$$\Delta = 0.50(0.95(12) - 0.80(12))$$

$$\Delta = 0.50(11.4 - 9.6)$$

$$\Delta = 0.50(1.8)$$

$$\Delta = 0.90 \text{ in}$$

$$\Delta = 0.50(0.95(12) - 0.80(12)) = 0.90 \text{ in}$$

2.7. Determine m

For annular strip behavior, conservatively assume:

$$m = \frac{B}{2}$$

Using:

$$B = 12 \text{ in}$$

Substitute:

$$m = \frac{12}{2}$$

$$m = 6 \text{ in}$$

$$m = \frac{12}{2} = 6 \text{ in}$$

2.8. Determine n

Governing Equation

$$n = \frac{B}{2}$$

$$B = 12 \text{ in}$$

Substitute:

$$m = \frac{12}{2}$$

$$m = 6 \text{ in}$$

Determine Required Plate Thickness

Use the larger value:

$$m = 6 \text{ in}$$

Governing Equation

$$t_p = (m \text{ or } n) \sqrt{\frac{2P_{strip}}{0.90F_yBN}}$$

$$t_p = 6 \sqrt{\frac{2(25.01)}{0.90(36)(12)(12)}}$$

$$t_p = 6 \sqrt{\frac{50.02}{4665.6}}$$

$$t_p = 6\sqrt{0.01072}$$

$$t_p = 6(0.1035)$$

$$t_p = 0.621 \text{ in}$$

$$t_p \approx 0.63 \text{ in}$$

$$\mathbf{t_p = 1 \text{ in}}$$

3.0. Design Against Moment:

Based on the AISC LRFD moment design procedure adapted for annular strip behavior.

3.1. Determine Factored Load and Moment

Given

$$P_u = 1650.2553 \text{ kips}$$

$$M_u = 4847.67 \text{ kip-ft}$$

Convert moment to kip-in:

$$M_u = 4847.67(12) = \mathbf{58172.04 \text{ kip-in}}$$

3.2. Determine Circumferential Moment per Unit Strip

For annular strip design, distribute moment around the circumference.

Governing Equation

$$M_{strip} = \frac{M_u}{\pi D_{cone}}$$

Substitute:

$$M_{strip} = \frac{58172.04}{\pi(21)}$$

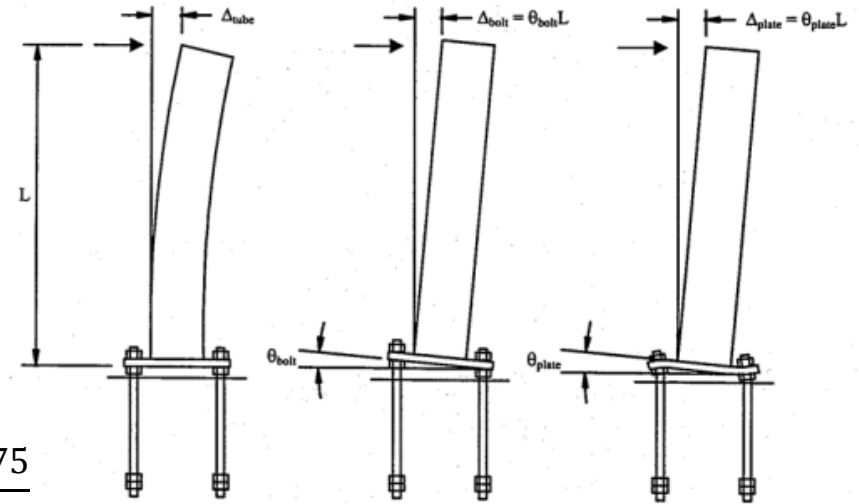
$$M_{strip} = \frac{58172.04}{65.973}$$

$$M_{strip} = 881.75 \text{ kip-in/ft}$$

Convert to kip-in/in:

$$M_{strip} = \frac{881.75}{12}$$

$$M_{strip} = 73.48 \text{ kip-in/in}$$



Components of Total Deflection cook el. 2000

Determine Equivalent Eccentricity

3.3.

Governing Equation

$$e = \frac{M}{P}$$

Using strip quantities:

$$e = \frac{73.48}{2.084}$$

$$e = 35.26 \text{ in}$$

$$e = \frac{73.48}{2.084} = 35.26 \text{ in}$$

3.4. Check Bearing Condition

For full compression:

$$e \leq \frac{N}{6}$$

Substitute:

$$\frac{N}{6} = \frac{12}{6}$$

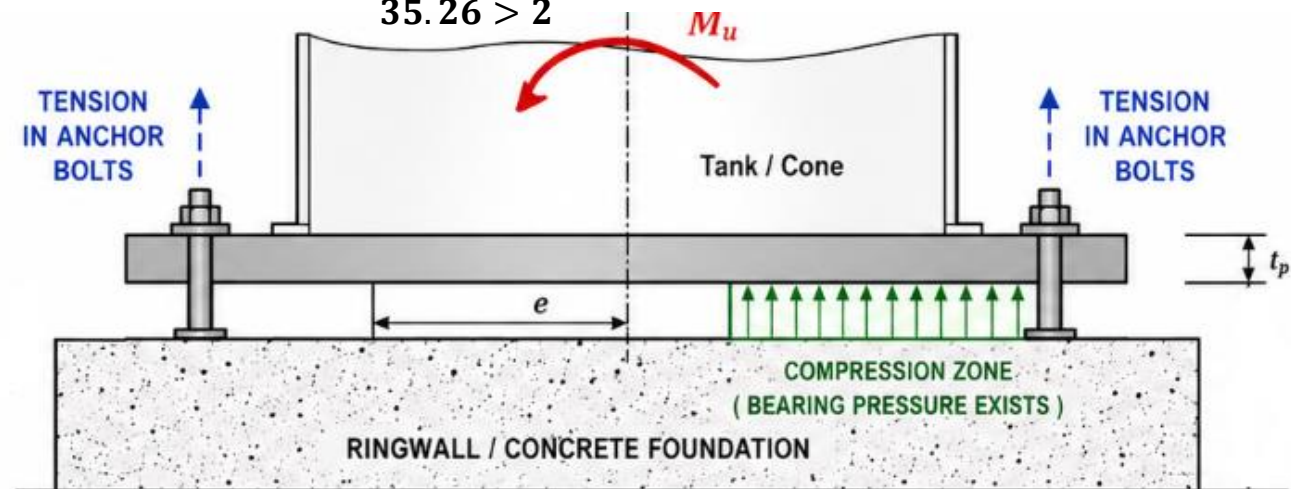
$$\frac{N}{6} = 2 \text{ in; Since:}$$

$$35.26 > 2$$

full compression does not occur.

Therefore:

- uplift develops,
- anchor bolts resist tension,
- partial compression exists beneath the annular plate.



$$e = 35.26 \text{ in} > \frac{12}{6} = 2 \text{ in}$$

3.5. Determine Critical Section

Using annular strip geometry:

$$m = \frac{B}{2}$$

Given:

$$B = 12 \text{ in}$$

Thus:

$$m = \frac{12}{2}$$

$$m = 6 \text{ in}$$

$$m = \frac{12}{2} = 6 \text{ in}$$

3.6. Determine Plastic Moment per 1-in Strip

Using the AISC LRFD moment procedure:

Governing Equation

$$t_p = \sqrt{\frac{4M_{plu}}{0.90F_y}}$$

$$t_p = \sqrt{\frac{4M_{plu}}{0.90F_y}}$$

3.7. Determine Strip Plastic Moment

Approximate factored strip moment:

$$M_{plu} = 73.48 \text{ kip-in/in}$$

3.8. Determine Required Thickness

Substitute:

$$t_p = \sqrt{\frac{4(73.48)}{0.90(36)}}$$

Denominator:

$$0.90(36) = 32.4$$

Numerator:

$$4(73.48) = 293.92$$

Thus:

$$t_p = \sqrt{\frac{293.92}{32.4}}$$

$$t_p = \sqrt{9.072}$$

$$t_p = 3.01 \text{ in}$$

$$t_p = \sqrt{\frac{4(73.48)}{0.90(36)}} = 3.01 \text{ in}$$

3.9. Final Thickness Selection

from axial load:

$$t_p = 0.63 \text{ in}$$

From moment loading:

$$t_p = 3.01 \text{ in}$$

Therefore, moment governs. **Adopt**

$$t_p = 3.0 \text{ in}$$

4.0. Local Uplift Bending Between Compression and Uplift Regions

the adopted thickness provides adequate resistance against:

- local uplift strip flexure,
- localized plate deformation,
- annular cantilever bending.

Therefore, local uplift bending is adequate.

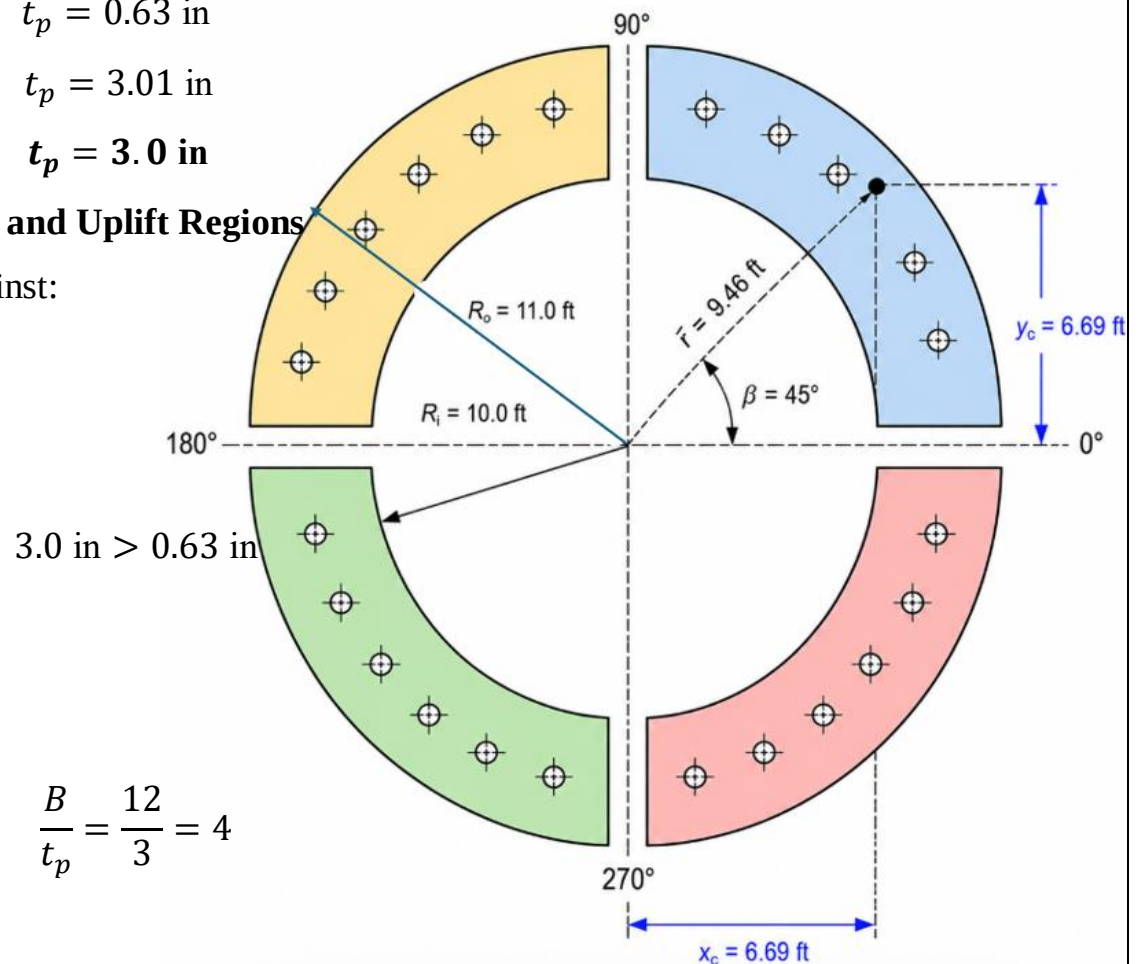
5.0. Local Plate Stability / Buckling Check

The compactness ratio of the annular strip is:

$$\frac{B}{t_p} = \frac{12}{3} = 4$$

$$\frac{B}{t_p} = \frac{12}{3} = 4$$

Because the plate is very stocky and compact:



- local plate buckling,
- compression instability,
- local flexural instability,

are not governing limit states.

Therefore, the annular plate is stable against local buckling.

6.0. Segment Angle

Since the annular base plate is divided into four equal segments:

$$\theta_{seg} = \frac{360^\circ}{4}$$

$$\theta_{seg} = 90^\circ$$

Convert to radians:

$$\theta_r = \frac{\pi}{2} = 1.5708 \text{ rad}$$

$$\theta_r = \frac{\pi}{2} = 1.5708 \text{ rad}$$

Plate Thickness

From overturning moment design:

$$t_p = 3.0 \text{ in}$$

Convert to feet:

$$t_p = \frac{3}{12} = 0.25 \text{ ft}$$

6.1. Centroid Radius of Annular Segment

Governing Equation

$$\bar{r} = \frac{4\sin(\theta_r/2)}{3\theta_r} \cdot \frac{R_o^3 - R_i^3}{R_o^2 - R_i^2}$$

Using:

$$R_o = 11.0 \text{ ft}$$

$$R_i = 10.0 \text{ ft}$$

$$\theta_r = 1.5708 \text{ rad}$$

Substituting:

$$\begin{aligned}\bar{r} &= \frac{4\sin(1.5708/2)}{3(1.5708)} \cdot \frac{11^3 - 10^3}{11^2 - 10^2} \\ &= \frac{4\sin(0.7854)}{4.7124} \cdot \frac{1331 - 1000}{121 - 100} \\ &= \frac{4(0.7071)}{4.7124} \cdot \frac{331}{21} \\ &= 0.6002(15.7619) \\ \bar{r} &= 9.46 \text{ ft}\end{aligned}$$

$$\bar{r} = 9.46 \text{ ft}$$

6.2 Centroid Coordinates

For Segment 1:

$$\beta = 45^\circ$$

x-Coordinate

$$x_c = \bar{r} \cos \beta$$

Substitute:

$$x_c = 9.46 \cos 45^\circ$$

$$x_c = 9.46(0.7071)$$

$$x_c = 6.69 \text{ ft}$$

$$x_c = 9.46 \cos 45^\circ = 6.69 \text{ ft}$$

y-Coordinate

$$y_c = \bar{r} \sin \beta$$

Substitute:

$$y_c = 9.46 \sin 45^\circ$$

$$y_c = 9.46(0.7071)$$

$$y_c = 6.69 \text{ ft}$$

$$y_c = 9.46 \sin 45^\circ = 6.69 \text{ ft}$$

6.3. Segment Centroid Coordinates

Segment	Angular Range	Centroid Angle β	$x_c = \bar{r} \cos \beta (\text{ft})$	$y_c = \bar{r} \sin \beta (\text{ft})$	Centroid Coordinates (ft)
Segment 1	$0^\circ \rightarrow 90^\circ$	45°	+6.69	+6.69	(6.69, 6.69)
Segment 2	$90^\circ \rightarrow 180^\circ$	135°	-6.69	+6.69	(-6.69, 6.69)
Segment 3	$180^\circ \rightarrow 270^\circ$	225°	-6.69	-6.69	(-6.69, -6.69)
Segment 4	$270^\circ \rightarrow 360^\circ$	315°	+6.69	-6.69	(6.69, -6.69)